Limits – 6

Asymptotics and sharp estimates: sums, integrals, Stirling

Homework

1 сентября 2026 года

Theoretical minimum. Comparison of a sum with an integral and Euler's constant; the Euler–Maclaurin formula and Bernoulli numbers; Stirling's formula and Wallis' product; asymptotics of partial sums and of tails of series; Stolz's rule as a way of extracting the leading term; the expansions of $\ln(1+x)$ and e^x as a source of corrections.

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1. Prove that $\sum_{k=n+1}^{\infty} \frac{1}{k^2} = \frac{1}{n} - \frac{1}{2n^2} + O\big(\frac{1}{n^3}\big)$ and find the analogous expansion for $\sum_{k>n} \frac{1}{k^3}$.

2. Find the asymptotics of $\big(1+\frac{1}{n}\big)^n$ up to $O(n^{-2})$ and deduce $\lim n\big(e - (1+\frac{1}{n})^n\big)$.

3. Prove that $\binom{2n}{n} = \frac{4^n}{\sqrt{\pi n}}\big(1 - \frac{1}{8n} + O\big(\frac{1}{n^2}\big)\big)$.

4. Assuming the prime number theorem in the form $\pi(x) = \frac{x}{\ln x}\big(1 + O\big(\frac{1}{\ln x}\big)\big)$, find the asymptotics of the n-th prime up to the leading term and explain where the correction in $p_n = n\ln n + n\ln\ln n + O(n)$ comes from.

5. Let $a_n = \int_0^{\pi/2} \sin^n x\,dx$. Find the asymptotics of a_n as $n \to \infty$ and prove that $n a_n a_{n-1} = \frac{\pi}{2}$.

6. Prove that $\sum_{k=1}^{n} \frac{1}{k} - \ln n \to \gamma$ and that $\gamma = \int_0^1 \big(\frac{1}{\ln x} + \frac{1}{1-x}\big)dx$.

7. Find $\lim_{n\to\infty} n\Big(\frac{\sum_{k=1}^{n} k^p}{n^{p+1}} - \frac{1}{p+1}\Big)$ for $p > 0$.

8. Prove that $\ln(n!) = n\ln n - n + \frac{1}{2}\ln n + C + o(1)$ by Stolz's method, without using the Euler–Maclaurin formula.

9. Let x_n be the root of $\tan x = x$ in the interval $\left(\pi n - \frac{\pi}{2}, \pi n + \frac{\pi}{2}\right)$. Find the asymptotics of x_n up to $O(n^{-3})$.

10. Prove that $\sum_{k=1}^{n} \frac{1}{\sqrt{k}} = 2\sqrt{n} + \zeta\left(\frac{1}{2}\right) + \frac{1}{2\sqrt{n}} + O\left(n^{-3/2}\right)$ and explain in what sense the value of the zeta function at $\frac{1}{2}$ appears here.